

Rye Stahle-Smith

mcryeserver@gmail.com | (802) 556-2005 | www.linkedin.com/in/rye-stahle-smith

EDUCATION

University of South Carolina

Columbia, SC

B.S.E. in Computer Engineering, GLD in Research, Minor in Mathematics

August 2023 – May 2026

Honors: NSPE-SC Outstanding Senior | Tau Sigma NHS Inductee | Cum Laude

GPA: 3.9

UNC at Asheville & North Carolina State University

Asheville & Raleigh, NC

Joint B.S. in Mechatronics Engineering

August 2021 – May 2023

Honors: Dean's List - Fall 2021 | Order of Omega NHS Inductee | Morehead Family Engineering Scholarship

RELATED EXPERIENCE

Swampfox Technologies | *Software Engineer* | Columbia, SC

May 2026 – Present

- Developed a strong understanding of company software platforms, system architectures, and deployment processes.
- Applied strong problem-solving and technical skills to support client implementations and resolve system issues.
- Collaborated with cross-functional teams to ensure efficient delivery, integration, and support of software solutions.
- Delivered memorable client experiences by providing clear communication, reliable solutions, and ongoing technical support.

University of South Carolina | *Research Assistant* | Columbia, SC

May 2025 – Present

- Conducted R&D in reconfigurable embedded systems and cloud-scale computing, specializing in hardware security, FPGA architectures, and real-time ML deployment.
- Developed BASE (*Byte-level Analytics and Statistical Engine*), a hardware accelerator for high-throughput statistical analysis with applications in security, machine learning, and data processing.
- Maintained and expanded BLADEI (*Bitstream-Level Abnormality Detection for Embedded Inference*) to support all PYNQ-compatible FPGA platforms; transitioned its ML pipeline from serialization to quantization to improve latency and portability.
- Managed concurrent research projects, experiments, technical documentation, and research deliverables while mentoring undergraduate researchers and facilitating knowledge transfer to new contributors.
- Presented research at the ACM Student Research Competition at SC25, UofSC Discover Day 2026, and IEEE SoutheastCon 2026 (accepted for publication and presentation).

Outlier | *AI Training Specialist* | Remote, USA

October 2024 – October 2025

- Evaluated CS-related prompts to train generative AI models, improving their coding efficiency and accuracy.
- Ensured AI solutions met the standards of clarity, correctness, and creativity, including solving math problems and verifying API calls.

Personal Programming Projects

Three-Stage Pipelined Processor

August 2024 – December 2024

- Designed and implemented a three-stage pipelined processor in Verilog, supporting a subset of the RV32I Base Integer Instruction Set for RISC-V, including R-, I-, U-, J-, and B-type instructions.
- Optimized the processor for efficient deployment on an FPGA board.

WORK EXPERIENCE

Motor Supply Company Bistro | *Server* | Columbia, SC

May 2024 – Present

- Gained in-depth familiarity with the restaurant's farm-to-table menu and beverage options.
- Enhanced teamwork and communication with colleagues to ensure smooth and efficient service operations.
- Created exceptional guest experiences through attentive service and a welcoming attitude.

Kiawah Island Golf Resort | *Server & Server Assistant* | Kiawah Island, SC

May 2023 – May 2024

- Provided customers with extensive knowledge of menu/beverage offerings showing strong interpersonal skills.
- Communicated effectively with staff, management, and the kitchen during service.
- Delivered memorable guest experiences through outstanding hospitality and efficient service.

Asheville Campus Entertainment | *Special Events Coordinator* | Asheville, NC

August 2022 – May 2023

- Orchestrated a diverse range of events to enhance the campus community experience.
- Organized events including venue selection, equipment procurement, & scheduling for seamless event execution.
- Engaged proactively with event-specific vendors to secure resources/services while negotiating favorable terms.

SKILLS

Software: Google Suite | Intel Quartus Prime | Vivado/Vitis | MATLAB/ Octave | R | UNIX/Linux | Git

Programming: Python | LC3/MIPS Assembly | Java | JEXL | C/C++/C# | Verilog | HTML/JS/TS | SCXML